

TOPIC 10.A

The Human Reproductive System and Family Planning

1. Describe the function of the ovary as an endocrine gland.
2. What regulates these processes?
3. Where and when does the egg cell (ovum) get fertilised naturally?
4. Knowing this, how can childbearing be planned?
5. What contraceptive methods do you know, and what are their risks?

Expected Knowledge & Marking Scheme

No	Expected Knowledge	Attainable Points	Attained points
1.	Ovary as a (paired) gonad: - Site of egg cell production. - From puberty, one egg matures in a cyclically repeating manner. As a hormone-producing organ: - The maturing follicle produces oestrogen. - High oestrogen levels trigger an increase in luteinising hormone. - The remains of the follicle transform into the corpus luteum, which produces progesterone.	8 pts	
2.	The rhythm of sexual functions is established by the hypothalamus. - Three hormones are produced in the anterior lobe of the pituitary gland that regulate gonad function: Follicle-Stimulating Hormone (FSH), Luteinising Hormone (LH), and Prolactin (Luteotropic hormone).	4 pts	
3.	Following ovulation (Day 14), the egg enters the fallopian tube. Sperm cells fertilise the egg cell here.	6 pts	
4.	Family planning can be managed using the calendar method. It is based on ensuring fertilisation occurs around the expected time of ovulation (usually days 10–17).	4 pts	
5.	- Coitus interruptus (withdrawal). - Condoms. - Hormonal contraception: Women take daily pills containing oestrogen and progesterone. - Intrauterine devices (IUDs): "The Coil." - Emergency contraception: "Morning-after" pills (high hormone content). - Termination of pregnancy (Abortion).	3 pts	

	Mentioning of risks: hormonal imbalance, intrauterine irritation etc.		
6.	Structure of the response, expressive skills	5 pts	
	Total	30 pts	

TOPIC 10.B

Measurement and Interpretation of Blood Pressure and Pulse

1. Using the provided device, measure your pulse and blood pressure (using an automated device).
2. Perform 20 squats, then stand up and measure your blood pressure again. How do the blood pressure and pulse values change?
3. Interpret the results of the measured data.

Expected Knowledge & Marking Scheme

No.	Expected Knowledge	Attainable points	Attained points
1.	Professional use of the equipment required for blood pressure and pulse measurement (with explanation).	5 pts	
2.	Both blood pressure and pulse rate will increase.	5 pts	
3.	Interpretation: Blood Pressure: - The hydrostatic pressure exerted by blood on the walls of the blood vessels. - We determine the highest and lowest values occurring in the arterial system of the systemic circulation. - The upper value (systole) is the pressure measured in the large arteries during the contraction of the left ventricle. - The lower value (diastole) is the value between two contractions of the left ventricle. - Automated measurement calculates these two values via an algorithm based on recording pressure fluctuations accompanying cardiac contraction and relaxation (oscillometric measurement). - Normal blood pressure value is 120/80 mmHg. - We can also distinguish between high (hypertension) and low (hypotension) blood pressure. Pulse: - The number of heart contractions per minute. Average: 60–75 bpm.	10 pts	
	Total	20 pts	